

MA1521 Calculus for Computing

Lecture Group 2

Lecturer: Prof To Wing Keung

Lesson 08: Friday 04 September 2026

(Week 4)

Remark. Q07.6 to Q07.9 are leftover exercises from the Tuesday class in Week 4.

Remark. Q08.1 to Q08.8 are on Section 13.7 of Thomas' Calculus (and Lecture 9 and Lecture 10 of Prof Auroux).



Q07.6. (i) Suppose $f(\mathbf{x}, \mathbf{y})$ is a function such that

$$f_x = 3x^2 + Axy, \quad f_y = 3x^2 + 6y^2,$$

where A is a constant. Using Clairaut's Theorem (i.e. the identity $f_{xy} = f_{yx}$), determine the value of A .

(ii) Using your result in (i), find such a function $f(\mathbf{x}, \mathbf{y})$ by inspection or any method using your experience in one variable calculus.

[Remark. Such a function is not unique.]

Solution: (i)

$$f_x = 3x^2 + Axy, \quad f_y = 3x^2 + 6y^2,$$

$$f_{xy} = (f_x)_y = \frac{\partial}{\partial y}(3x^2 + Axy) = 0 + Ax = \underline{Ax}.$$

$$f_{yx} = (f_y)_x = \frac{\partial}{\partial x}(3x^2 + 6y^2) = 6x + 0 = \underline{6x}.$$

By Clairaut's Theorem, we have

$$f_{xy} = f_{yx} \implies Ax = 6x \implies \underline{A = 6}$$

upon evaluating both sides at some $x \neq 0$.

(ii) By (i), we have $\mathbf{A} = \mathbf{6}$, and thus

$$f_x = 3x^2 + 6xy, \quad f_y = 3x^2 + 6y^2,$$

$$\begin{aligned} f(x,y) &= \int f_x \, dx \\ &= \int (3x^2 + 6xy) \, dx \end{aligned}$$

$$f(x,y) = x^3 + 3x^2y + \phi(y)$$

Take $\frac{\partial}{\partial y}$ on both sides,

↪ does not depend on x ,
but may depend on y .

$$f_y = 0 + 3x^2 + \phi'(y)$$

$$\cancel{3x^2} + 6y^2 = \cancel{3x^2} + \phi'(y) \Rightarrow \phi'(y) = 6y^2$$

$$\Rightarrow \phi(y) = \int \phi'(y) \, dy = \int 6y^2 \, dy = 2y^3 + C$$

$$\therefore f(x,y) = x^3 + 3x^2y + 2y^3 + C$$

Q07.7. Let $f(\mathbf{x}, y)$ be a function, and $(\mathbf{x}_o, y_o)$ be a point in the domain of f . Consider the linear polynomial (in two variables) given by

$$g(\mathbf{x}, y) = \mathbf{a} + \mathbf{b}(\mathbf{x} - \mathbf{x}_o) + c(y - y_o)$$

where $\mathbf{a}, \mathbf{b}, c$ are constants.

(i) Determine the values of $\mathbf{a}, \mathbf{b}, c$ (in terms of data on f) if

$$g(\mathbf{x}_o, y_o) = f(\mathbf{x}_o, y_o), \quad g_{\mathbf{x}}(\mathbf{x}_o, y_o) = f_{\mathbf{x}}(\mathbf{x}_o, y_o)$$

$$\text{and} \quad g_y(\mathbf{x}_o, y_o) = f_y(\mathbf{x}_o, y_o).$$

(ii) Write down the equation for the graph of $g(\mathbf{x}, y)$.

[Remark: Note that the equation in (ii) is simply an equation of the tangent plane to the graph of $f(\mathbf{x}, y)$ at the point $(\mathbf{x}_o, y_o)$ as given in Prof Auroux's notes, Lecture 9.]

$$\mathbf{g}(\mathbf{x}_o, y_o) = \mathbf{f}(\mathbf{x}_o, y_o), \quad \mathbf{g}_x(\mathbf{x}_o, y_o) = \mathbf{f}_x(\mathbf{x}_o, y_o)$$

and $\mathbf{g}_y(\mathbf{x}_o, y_o) = \mathbf{f}_y(\mathbf{x}_o, y_o).$ (*)

Solution: (i)

$$\mathbf{g}(\mathbf{x}, y) = \mathbf{a} + \mathbf{b}(\mathbf{x} - \mathbf{x}_o) + \mathbf{c}(y - y_o)$$

$$\implies \mathbf{g}_x = \mathbf{b}, \quad \mathbf{g}_y = \mathbf{c}.$$

Note also that $\mathbf{g}(\mathbf{x}_o, y_o) = \mathbf{a}$. Together with (*), we have

$$\mathbf{g}(\mathbf{x}_o, y_o) = \mathbf{a} = \mathbf{f}(\mathbf{x}_o, y_o) \implies \mathbf{a} = \mathbf{f}(\mathbf{x}_o, y_o),$$

$$\mathbf{g}_x(\mathbf{x}_o, y_o) = \mathbf{b} = \mathbf{f}_x(\mathbf{x}_o, y_o) \implies \mathbf{b} = \mathbf{f}_x(\mathbf{x}_o, y_o),$$

$$\mathbf{g}_y(\mathbf{x}_o, y_o) = \mathbf{c} = \mathbf{f}_y(\mathbf{x}_o, y_o) \implies \mathbf{c} = \mathbf{f}_y(\mathbf{x}_o, y_o).$$

Thus we have

$$\mathbf{g}(\mathbf{x}, y) = \mathbf{f}(\mathbf{x}_o, y_o) + \mathbf{f}_x(\mathbf{x}_o, y_o)(\mathbf{x} - \mathbf{x}_o) + \mathbf{f}_y(\mathbf{x}_o, y_o)(y - y_o).$$

(ii) By (i), we have

$$g(\mathbf{x}, \mathbf{y}) = f(\mathbf{x}_o, \mathbf{y}_o) + f_x(\mathbf{x}_o, \mathbf{y}_o)(\mathbf{x} - \mathbf{x}_o) + f_y(\mathbf{x}_o, \mathbf{y}_o)(\mathbf{y} - \mathbf{y}_o).$$

Thus graph of g is given by the equation

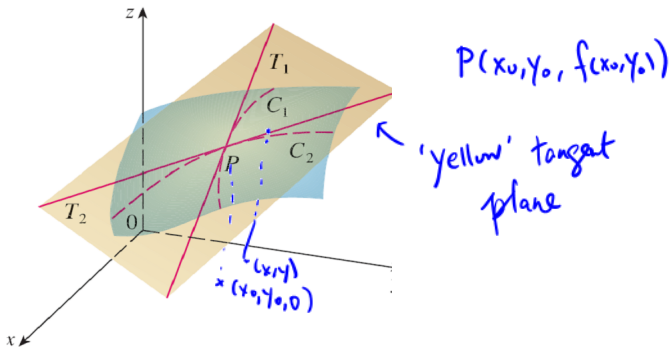
$$z = f(\mathbf{x}_o, \mathbf{y}_o) + f_x(\mathbf{x}_o, \mathbf{y}_o)(\mathbf{x} - \mathbf{x}_o) + f_y(\mathbf{x}_o, \mathbf{y}_o)(\mathbf{y} - \mathbf{y}_o).$$

Remark. Note that the equation in (ii) is simply an equation of the tangent plane to the graph of $f(\mathbf{x}, \mathbf{y})$ at the point $(\mathbf{x}_o, \mathbf{y}_o)$ as given in Prof Auroux's notes, Lecture 9.]

Remark. $g(\mathbf{x}, \mathbf{y})$ (or equivalently, the z -value of the tangent plane) gives the linear approximation of $f(\mathbf{x}, \mathbf{y})$ for points $(\mathbf{x}, \mathbf{y})$ near $(\mathbf{x}_o, \mathbf{y}_o)$.

Notation as in Tuesday's slides.

The tangent plane to the surface $\mathbf{S}$ (the graph $\mathbf{z} = \mathbf{f}(\mathbf{x}, \mathbf{y})$) at the point $\mathbf{P}(\mathbf{a}, \mathbf{b}, \mathbf{f}(\mathbf{a}, \mathbf{b}))$ is the plane that contains both the tangent lines $\mathbf{T}_1$ of the curve $\mathbf{C}_1$ at $\mathbf{P}$ and the tangent line $\mathbf{T}_2$ of $\mathbf{C}_2$ at $\mathbf{P}$.



Q07.8. Consider the function $f(x, y) = x^2y + 2xy$. Find an equation of the tangent plane to the graph of $f(x, y)$ at the point $(1, 2)$. [Recall the result in **Q07.7**.]

Solution:

$$f(x, y) = x^2y + 2xy \qquad f(1, 2) = 2 + 4 = 6$$
$$\Rightarrow f_x(x, y) = 2xy + 2y, \quad f_y(x, y) = x^2 + 2x$$

$$\therefore f_x(1, 2) = 4 + 4 = 8, \quad f_y(1, 2) = 1 + 2 = 3$$

$\therefore$ an equation of the tangent plane to the graph of f at $(1, 2)$ is:

$$z = f(1, 2) + f_x(1, 2) \cdot (x - 1) + f_y(1, 2) \cdot (y - 2)$$

$$z = 6 + 8(x - 1) + 3(y - 2).$$

Recall the tangent plane to the graph of
 $f(x, y) = x^2y + 2xy$ at $(1, 2)$:

$$z = 6 + 8(\overset{x}{\cancel{x}} - 1) + 3(y - 2).$$

An approximation of $f(1.05, 2.1)$ is given by the z -value of the above tangent plane. In other words, we have

$$f(1.05, 2.1) \approx 6 + 8(1.05 - 1) + 3(2.1 - 2) = 6.70.$$

(Actually,

$$f(1.05, 2.1) = (1.05)^2 \cdot (2.1) + 2 \cdot 1.05 \cdot 2.1 \approx 6.73)$$

Linear approximation. In the notation of Q07.7, for points $(\mathbf{x}, \mathbf{y})$ near $(\mathbf{x}_o, \mathbf{y}_o)$, we get an approximation of value of $f(\mathbf{x}, \mathbf{y})$ by $g(\mathbf{x}, \mathbf{y})$. In other words, we have

$$f(\mathbf{x}, \mathbf{y}) \approx f(\mathbf{x}_o, \mathbf{y}_o) + f_x(\mathbf{x}_o, \mathbf{y}_o)(\mathbf{x} - \mathbf{x}_o) + f_y(\mathbf{x}_o, \mathbf{y}_o)(\mathbf{y} - \mathbf{y}_o).$$

Write

$$\Delta \mathbf{x} = \mathbf{x} - \mathbf{x}_o, \quad \Delta \mathbf{y} = \mathbf{y} - \mathbf{y}_o, \quad \Delta f = f(\mathbf{x}, \mathbf{y}) - f(\mathbf{x}_o, \mathbf{y}_o).$$

Then the above approximation becomes

$$\Delta f \approx f_x(\mathbf{x}_o, \mathbf{y}_o)\Delta \mathbf{x} + f_y(\mathbf{x}_o, \mathbf{y}_o)\Delta \mathbf{y}.$$

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{\Delta f}{\Delta x}$$

$$\Delta f \approx f_x(x_o, y_o)\Delta x + f_y(x_o, y_o)\Delta y.$$

Roughly speaking, the function $f(x, y)$ is **differentiable** at (x_o, y_o) means that the above is a good approximation, meaning that as Δx and $\Delta y \rightarrow 0$, the error in the above approximation tends to **0** much faster than Δx and Δy . (See Thomas' Calculus, p.796).

In MA1521, we will focus on differentiable functions.

Q07.9. Consider the surface in $\mathbb{R}^3$ defined by the equation $yz = x^2$. Find an equation of the tangent plane to the surface at the point $(4, 2, 8)$.

[Suggestion: First rewrite the surface as the graph of a function of two variables.]

$$yz = x^2 \Rightarrow z = \frac{x^2}{y} \qquad f(x, y) = \frac{x^2}{y}$$

Given a point $(4, 2, 8)$ on the surface $yz = x^2$.

Check: Note that $2 \cdot 8 = 4^2$. Thus $(4, 2, 8)$ is indeed a point on the surface.

Solution: Note that $yz = x^2 \implies \underline{z = \frac{x^2}{y}}$. Thus we may write the surface as the graph of the function $\frac{x^2}{y}$ (at least near the point $(x, y) = \underline{(4, 2)}$).

Denote this function simply as $z = z(x, y) = \frac{x^2}{y}$. Then

$$\frac{\partial z}{\partial x} = \frac{2x}{y}, \quad \frac{\partial z}{\partial y} = -\frac{x^2}{y^2}.$$

Thus

$$\frac{\partial z}{\partial x}(4, 2) = \frac{2 \cdot 4}{2} = 4, \quad \frac{\partial z}{\partial y}(4, 2) = -\frac{4^2}{2^2} = -4.$$

Using the data

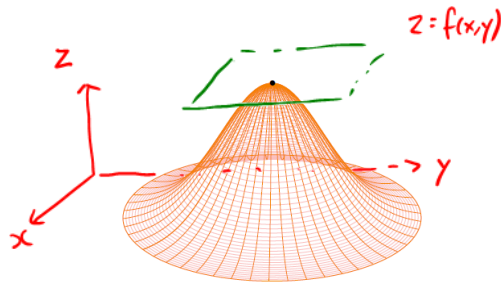
$$z(4, 2) = 8, \quad \frac{\partial z}{\partial x}(4, 2) = 4, \quad \frac{\partial z}{\partial y}(4, 2) = -4,$$

an equation of the tangent plane to the given surface at the point $(4, 2, 8)$ is:

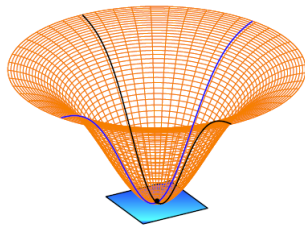
$$z = 8 + 4(x - 4) - 4(y - 2)$$

or equivalently,

$$z = 4x - 4y.$$



local maximum



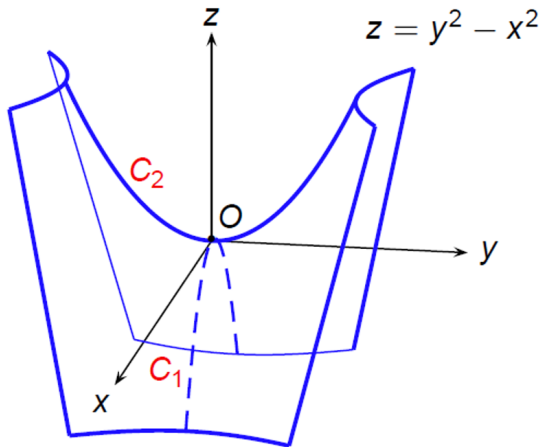
local minimum

Similar to the one variable case, the **critical points** of a differentiable function $f(x, y)$ are those points (a, b) such that

$$f_x(a, b) = 0 \quad \text{and} \quad f_y(a, b) = 0.$$

Remark. If a function $f(x, y)$ has a local maximum or local minimum point at (a, b) , then $f(x, y)$ also has a critical point at (a, b) .

The converse does not hold. In other words, there may be critical points of $f(x, y)$ which are not local max or local min points of $f(x, y)$. Such points are called **saddle points** of $f(x, y)$.

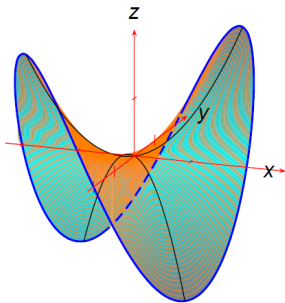


saddle point

Suppose you are standing on a surface and you are standing upright (parallel to the $\mathbf{z}$ -axis).

Moreover, when you begin walking, some directions take you uphill while other directions take you downhill.

Then you are standing at a saddle point!



Q08.1. Find the critical points of the function

$$f(x, y) = 2xy - x^2 - 2y^2 + 3x.$$

$$f_x = 2y - 2x + 3, \quad f_y = 2x - 4y$$

To find critical points of f , we solve

$$\begin{cases} f_x(x, y) = 0 \\ f_y(x, y) = 0 \end{cases} \Leftrightarrow \begin{cases} 2y - 2x + 3 = 0 & \text{---(1)} \\ 2x - 4y = 0 & \text{---(2)} \end{cases}$$

$$(1) + (2) \text{ gives } -2y + 3 = 0 \Rightarrow y = \frac{3}{2}$$

$$\text{From (2), } x = 2y = 2 \cdot \frac{3}{2} = 3.$$

$\therefore f$ has one critical point at $(3, \frac{3}{2})$.

Second derivative test:

Suppose $f_x(a, b) = f_y(a, b) = 0$.

Define the **discriminant** $D(a, b)$ for the point (a, b) by

$$\begin{aligned} D = D(a, b) &= \begin{vmatrix} f_{xx}(a, b) & f_{xy}(a, b) \\ f_{xy}(a, b) & f_{yy}(a, b) \end{vmatrix} \\ &= f_{xx}(a, b)f_{yy}(a, b) - [f_{xy}(a, b)]^2. \end{aligned}$$

(a) If $D > 0$ and ($f_{xx}(a, b) > 0$ or $f_{yy}(a, b) > 0$), then $f(a, b)$ is a local minimum.

→ (b) If $D > 0$ and ($f_{xx}(a, b) < 0$ or $f_{yy}(a, b) < 0$), then $f(a, b)$ is a local maximum.

(c) If $D < 0$, then (a, b) is a saddle point of f .

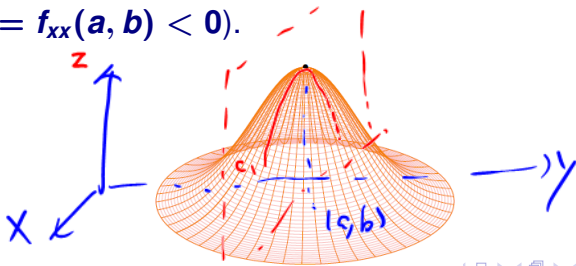
(d) If $D = 0$, then no conclusion can be drawn.

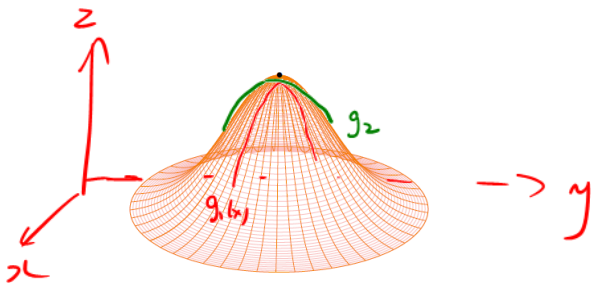
A glimpse on why the second derivative test is valid:

Take the special case where $f_{xy}(a, b) = 0$, so that $D = f_{xx}(a, b)f_{yy}(a, b)$.

Say, in case (b), if $D > 0$ and $f_{xx}(a, b) < 0$, then we must also have $f_{yy}(a, b) < 0$.

By restricting the domain of $f(x, y)$ to the line $y = b$, then the one variable function $g_1(x) = f(x, b)$ will have a local max at $x = a$ (since $g_1'(a) = f_x(a, b) = 0$ and $g_1''(a) = f_{xx}(a, b) < 0$).





Similarly, the function $\mathbf{g_2(y) = f(a, y)}$ will also have a local max at $\mathbf{y = b}$.

Thus $\mathbf{f(x, y)}$ has a local max at $\mathbf{(a, b)}$ along the $\mathbf{x}$ -direction and along the $\mathbf{y}$ -direction.

If $f(\mathbf{x}, \mathbf{y})$ is nice enough, the behaviour of $f(\mathbf{x}, \mathbf{y})$ along the $\mathbf{x}$ -direction and the $\mathbf{y}$ -direction will determine the behaviour of $f(\mathbf{x}, \mathbf{y})$ along other directions as well.

In particular, $f(\mathbf{x}, \mathbf{y})$ will have a local max at $(\mathbf{a}, \mathbf{b})$ along all directions. Hence $f(\mathbf{x}, \mathbf{y})$ will indeed have a local max at $(\mathbf{a}, \mathbf{b})$.

It turns out that the above special case is actually not so special. In fact, for any function $f(\mathbf{x}, \mathbf{y})$, one can always make a rotation of the $\mathbf{x}$ -axis and the $\mathbf{y}$ -axis, so that with respect to the new coordinates, f will take the form of the special case.

Q08.2. Determine all the critical points of the function

$$f(x, y) = 2xy - x^2y - xy^2,$$

and classify each critical point as a local maximum point, a local minimum point or a saddle point of f . Evaluate also the local maximum values and local minimum values of f , if any.

[Suggestion: You may find the second derivative test useful.]

Solution: Given $f(x, y) = 2xy - x^2y - xy^2$. We have

$$f_x = 2y - 2xy - y^2, \quad f_y = 2x - x^2 - 2xy,$$

$$f_{xx} = -2y, \quad f_{yy} = -2x, \quad f_{xy} = 2 - 2x - 2y.$$

To find critical points of f , we solve

$$\begin{cases} f_x(x, y) = 0 \\ f_y(x, y) = 0 \end{cases} \iff \begin{cases} 2y - 2xy - y^2 = 0, & (1) \\ 2x - x^2 - 2xy = 0. & (2) \end{cases}$$

From (1), we get

$$y(2 - 2x - y) = 0 \implies y = 0 \text{ or } y = 2 - 2x.$$

Substituting $y = 0$ into (2), we get

$$\begin{aligned} 2x - x^2 - 0 &= 0 \implies x(2 - x) = 0 \\ \implies x &= 0 \text{ or } x = 2 \implies (x, y) = (0, 0) \text{ or } (2, 0). \end{aligned}$$

Substituting $y = 2 - 2x$ into (2), we get

$$2x - x^2 - 2x(2 - 2x) = 0$$

$$\implies x(-2 + 3x) = 0$$

$$\implies x = 0 \text{ or } x = \frac{2}{3}$$

$$\implies (x, y) = (0, 2 - 2 \cdot 0) = (0, 2) \text{ or}$$

$$(x, y) = \left(\frac{2}{3}, 2 - 2 \cdot \frac{2}{3}\right) = \left(\frac{2}{3}, \frac{2}{3}\right).$$

Thus the critical points of f are $(0, 0)$, $(2, 0)$, $(0, 2)$ and $\left(\frac{2}{3}, \frac{2}{3}\right)$.

Now

$$D(x, y) = f_{xx} \cdot f_{yy} - (f_{xy})^2 = 4xy - (2 - 2x - 2y)^2.$$

It follows that

$$\left\{ \begin{array}{l} D(0, 0) = 0 - 2^2 = -4 < 0, \\ D(2, 0) = 0 - (2 - 4 - 0)^2 = -4 < 0, \\ D(0, 2) = 0 - (2 - 0 - 4)^2 = -4 < 0, \end{array} \right.$$

$$D\left(\frac{2}{3}, \frac{2}{3}\right) = \frac{16}{9} - \frac{4}{9} = \frac{4}{3} > 0,$$

- 2y

$$\text{and } f_{xx}\left(\frac{2}{3}, \frac{2}{3}\right) = -2 \cdot \frac{2}{3} = -\frac{4}{3} < 0.$$

Thus by the second derivative test, f has a saddle point at

$(0, 0)$, $(2, 0)$ and $(0, 2)$; and f has a local maximum at

$$\left(\frac{2}{3}, \frac{2}{3}\right) \text{ with value } f\left(\frac{2}{3}, \frac{2}{3}\right) = \frac{2}{3} \cdot \frac{2}{3} \cdot \left(2 - \frac{2}{3} - \frac{2}{3}\right) = \frac{8}{27}.$$

Q08.3. Determine all the critical points of the function

$$f(x, y) = \ln(xy^2) - xy - 2y$$

over the region $x > 0$, $y > 0$ in the xy -plane. Classify each critical point as a local maximum point, a local minimum point or a saddle point of f . Evaluate also the local maximum values and local minimum values of f , if any.

[Suggestion: You may find the second derivative test useful.]

Solution: (i) Let

$$\begin{aligned} f(x, y) &= \ln(xy^2) - xy - 2y \\ &= \ln x + 2 \ln y - xy - 2y, \quad \text{where } x > 0, y > 0. \end{aligned}$$

We have $f_x = \frac{1}{x} - y$ and $f_y = \frac{2}{y} - x - 2$.

Also, $f_{xx} = -\frac{1}{x^2}$, $f_{xy} = -1$ and $f_{yy} = -\frac{2}{y^2}$.

To find critical points of f , we solve

$$\begin{cases} f_x(x, y) = 0 \\ f_y(x, y) = 0 \end{cases} \iff \begin{cases} \frac{1}{x} - y = 0, & (1) \\ \frac{2}{y} - x - 2 = 0. & (2) \end{cases}$$

From (1), we get $y = \frac{1}{x}$. (3)

Substituting (3) into (2), we get

$$2x - x - 2 = 0 \implies x = 2 \implies y = \frac{1}{2}.$$

Thus, the only critical point of f is $(2, \frac{1}{2})$. Now

$$\begin{aligned} D(2, \frac{1}{2}) &= f_{xx}(2, \frac{1}{2}) \cdot f_{yy}(2, \frac{1}{2}) - [f_{xy}(2, \frac{1}{2})]^2 \\ &= (-\frac{1}{4}) \cdot (-8) - (-1)^2 = \underline{1} > 0, \quad \text{and} \end{aligned}$$

$$\underline{f_{xx}(2, \frac{1}{2})} = \underline{-\frac{1}{4}} < 0.$$

Thus by the second derivative test, f has a local maximum at $(2, \frac{1}{2})$ with value

$$f(2, \frac{1}{2}) = \ln\left(\frac{1}{2}\right) - 1 - 1 = -\ln 2 - 2.$$

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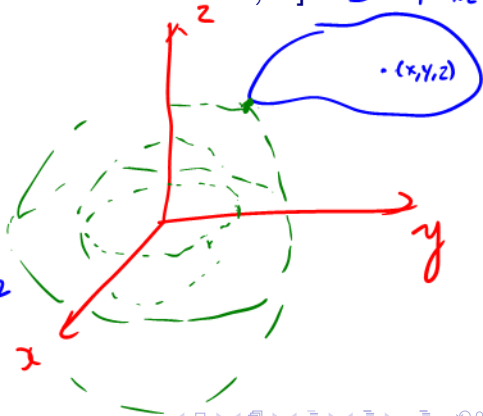
Q08.4. Consider the surface $y^2 - xz = 4$ in the xyz -space. Find the point(s) on the surface which is closest to the origin $(0, 0, 0)$.

[Suggestion: First formulate the question as a problem to minimize a certain function in the variables x, z .] S $y^2 - xz = 4$

Distance = $\sqrt{x^2 + y^2 + z^2}$.
from $(0, 0, 0)$

$$x^2 + y^2 + z^2$$

$$f(x, z) = x^2 + (4 + xz) + z^2$$



Solution: We would like to minimize $\sqrt{x^2 + y^2 + z^2}$ or equivalently, minimize

$$x^2 + y^2 + z^2 \quad (1)$$

subject to the constraint $y^2 - xz = 4$. (2)

Substituting (2) into (1), the problem becomes:

To minimize the function $f(x, z) = x^2 + 4 + xz + z^2$.

$$f_x = 2x + z, \quad f_z = x + 2z.$$

To find critical points of f , we solve

$$\begin{cases} f_x(x, z) = 0 \\ f_z(x, z) = 0 \end{cases} \iff \begin{cases} 2x + z = 0, & (3) \\ x + 2z = 0. & (4) \end{cases}$$

From (2), we have $x = -2z$. Substituting this into (1), we get

$$2(-2z) + z = 0 \implies -3z = 0 \implies z = 0.$$

Then $x = -2 \cdot 0 = 0$. Thus $f(x, z)$ has one critical point at $(x, z) = (0, 0)$.

Substituting $x = z = 0$ into (2), we get

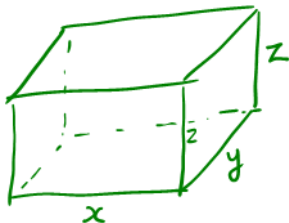
$$y^2 - 0 \cdot 0 = 4 \implies y^2 = 4 \implies y = 2, -2.$$

These correspond to the two points $(0, 2, 0)$ and $(0, -2, 0)$ on the surface. Note that both of these two points are at a distance of $\sqrt{0^2 + (\pm 2)^2 + 0^2} = 2$ from $(0, 0, 0)$.

So the points on the surface closest to $(0, 0, 0)$ are $(0, 2, 0)$ and $(0, -2, 0)$.

Q08.5. A rectangular box with its top left open has its five faces made from different materials. The front face and the back face are made from materials costing 10 dollars per square meter. The two lateral faces are made from materials costing 5 dollars per square meter. The bottom face is made from materials costing 30 dollars per square meter. Find the minimum material cost of the rectangular box if the volume of the box is 6 cubic meters. Give also the dimensions (i.e. length, width, and height) of such box.

Solution: Let x , y , z be the length, width and height (in meters) of the rectangular box, and let f be the corresponding material cost (in dollars) of the box. Then



$$f = 2 \cdot 10 \cdot xz + 2 \cdot 5 \cdot yz + 30 \cdot xy. \quad (1)$$

Since volume of the box is 6, we have the constraint

$$xyz = 6 \implies z = \frac{6}{xy}. \quad (2)$$

Substituting (2) into (1), we write f as a function of x , y :

$$f(x, y) = 20x \cdot \frac{6}{xy} + 10y \cdot \frac{6}{xy} + 30xy = \frac{120}{y} + \frac{60}{x} + 30xy.$$

$$f(x, y) = \frac{120}{y} + \frac{60}{x} + 30xy. \quad (3)$$

$$\text{Then } f_x = -\frac{60}{x^2} + 30y, \quad f_y = -\frac{120}{y^2} + 30x.$$

To find critical points of f , we solve

$$\begin{cases} f_x(x, y) = 0 \\ f_y(x, y) = 0 \end{cases} \iff \begin{cases} -\frac{60}{x^2} + 30y = 0, & (4) \\ -\frac{120}{y^2} + 30x = 0. & (5) \end{cases}$$

From (4), we have $y = \frac{2}{x^2}$. Substituting this into (5), we get

$$-\frac{120}{(2/x^2)^2} + 30x = 0 \Rightarrow -x^4 + x = 0 \Rightarrow x(1 - x^3) = 0 \Rightarrow x = 1, 0.$$

Due to (2), $x = 0$ is rejected. So $x = 1$.

Then $y = \frac{2}{1^2} = 2$. Together with (2), we have

$$z = \frac{6}{1 \cdot 2} = 3.$$

So the desired box is of length=1 m, width=2 m, and height=3 m, and the minimum material cost is

$$f(1, 2) = \frac{120}{2} + \frac{60}{1} + 30 \cdot 1 \cdot 2 = 180 \text{ dollars.}$$